

NOT ON YOUR FORMULA SHEET		
TOPIC	THE RULE YOU MUST BRING YOURSELF	IN FULL
Indices	$1/x^n = x^{-n}$ $x^{0.5} = \sqrt{x}$ $x^0 = 1$	$\sqrt[n]{x} = x^{1/n}$ DIFFERENTIATE
Logs	change of base $\log_b x = \frac{\lg x}{\lg b}$ the swap $y = b^x$ means $x = \log_b y$	LOGS
Lines	$y - y_1 = m(x - x_1)$ parallel $m_1 = m_2$, perpendicular $m_1 m_2 = -1$, collinear = same gradient	STRAIGHT LINES
Quadratics	$b^2 - 4ac$: above 0 two roots, 0 one, below 0 none a positive gives a MINIMUM, a negative gives a MAXIMUM	QUADRATICS
Factor theorem	$f(a) = 0$ means $(x - a)$ is a factor	PARTIAL FRACTIONS
Triangles	SOH CAH TOA sine rule $\frac{\sin A}{a} = \frac{\sin B}{b}$ cosine rule $a^2 = b^2 + c^2 - 2bc \cos A$	TRIG
Vectors	magnitude $ v = \sqrt{x^2 + y^2}$, $AB = B - A$ which quarter the angle sits in	VECTORS
Matrices	order is rows $\times$ columns, and the inside numbers must match	MATRICES
Differentiate	the chain rule, and $d/dx \ln f = f'/f$ $d/dx A \sin(Bx+C) = AB \cos(Bx+C)$	DIFFERENTIATE
Turning points	2nd derivative positive = MINIMUM, negative = MAXIMUM	TURNING POINTS
Integrate	$\int a/(bx+d) = (a/b) \ln bx+d + c$ below the axis: split there, drop the minus, then add	INTEGRATE
Waves	everything about $y = A \sin(B\theta+C) + D$ except the identities	WAVES
Diff. equations	all three equation-and-solution pairs	FIRST ORDER

STUCK? WHAT IS IT ASKING FOR		
THE QUESTION SAYS	IT WANTS	GO TO
"dy/dx", "rate of change", "gradient function"	differentiate	DIFFERENTIATE
"maximum", "minimum", "turning point"	differentiate, =0, sub into ORIGINAL	TURNING POINTS
"area under", numbers on $\int$	integrate, top take bottom, no +c	INTEGRATE
"roots", "crosses the x-axis"	set $y = 0$	QUADRATICS
"y-intercept"	it IS c	free mark
"in terms of x"	that letter alone on the left	REARRANGING
"show your working"	every line earns; a bare wrong answer earns nothing	always write line 1

QUADRATICS		
ASKED FOR	MOVE	WORKED
y-intercept	it is c	$6x^2 + 14x - 12 \rightarrow -12$
roots, factorises	two numbers $\times$ to c, + to b	$(3x-2)(x+3) \rightarrow 2/3, -3$
roots, does not	name a,b,c with signs, sheet formula	$196+288=484, \sqrt{484}=22$
max or min	sign of a only. +ve MIN, -ve MAX	$a=-6 \rightarrow$ maximum
how many roots	b^2-4ac : >0 two, =0 one, <0 none	$16-32=-16 \rightarrow$ none
line of symmetry	$x = -b/2a$. write it as x = ...	roots 3, 5 $\rightarrow x = 4$
turning point	that x, then sub into the ORIGINAL	$\rightarrow (4, -1)$
$(x+q)^2+r$	$q = b/2, r = c - q^2$	$x^2-10x+3 \rightarrow (x-5)^2-22$
$p(x+q)^2+r, a \neq 1$	$q = b/2a, r = c - b^2/4a$. do NOT halve b first	$2x^2+8x+15 \rightarrow 2(x+2)^2+7$
given as $f(x)$	EXPAND before reading a, b, c	$(3x-2)(4-2x) \rightarrow a=-6$
y not alone	rearrange to $y = \dots$ first	$y+19x-x^2=10x-10 \rightarrow x^2-9x-10$
"height after 2 s"	just substitute	$6+260-20 = 246$ m
"max height"	$x = -b/2a$, then sub back	$x=13 \rightarrow 851$ m
"hits the ground"	set $h = 0$, discard negative time	$\rightarrow 2.21$ s

QUADRATICS

ASKED FOR

MOVE

WORKED

roots are where it crosses. The y-intercept is c. The turning point sits on $x = -b/2a$.

STRAIGHT LINES AND COORDINATE GEOMETRY		
GIVEN / ASKED	MOVE	WORKED
two points $\rightarrow$ gradient	up over along, SAME order top and bottom	$(2,0), (4,10) \rightarrow 5$
equation not tidy	divide EVERY term into $y = mx+c$ first	$3y=6x+24 \rightarrow y=2x+8$
x-intercept	set $y = 0$	$0=2x+8 \rightarrow -4$
gradient + a point	$y-y_1 = m(x-x_1)$	$m=4, (6,-2) \rightarrow y=4x-26$
two points, no gradient	gradient first, then as above	$(2,4), (5,-2) \rightarrow y=-2x+8$
parallel?	same m. rearrange each one first	$6x-2y=10 \rightarrow m=3 \checkmark$
perpendicular / normal	$m_2 = -1/m_1$ flip and change sign	$m=4/5 \rightarrow -5/4$
collinear?	gradient PQ = gradient QR	all $-1/2 \rightarrow$ yes
perpendicular bisector	midpoint AND $-1/m$. both halves	$A(4,3)B(-2,5) \rightarrow y=3x+1$
right-angled triangle?	two gradients multiply to -1	$(4/3)(-3/4) = -1 \checkmark$
area of a triangle	$\frac{1}{2} x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)$	$\rightarrow 10$
sketch a region	dotted for $<$, solid for $\geq$. test (0,0)	true $\rightarrow$ shade that side

LOGS AND EXPONENTIALS		
LOOKS LIKE	MOVE	WORKED
$\log_b 8 = 3$	read aloud: what power of b gives 8	$b^3=8 \rightarrow b = 2$
$\log_a b = 2$	the answer IS the power	$4^2 \rightarrow b = 16$
$\log_2(\log_3 x)=1$	peel the OUTSIDE first	$\log_3 x=2 \rightarrow x = 9$
$\lg x = \lg 3 + \lg 8$	added $\rightarrow$ insides multiply	$x = 24$
$\lg x = \lg 40 - \lg 5$	subtracted $\rightarrow$ insides divide	$x = 8$
$\log_2 1 + \log_5 4$	$\log 1 = 0$; power comes to the front	$0+4 = 4$
$\log_2 10$, no button	change of base: $\lg 10 / \lg 2$	$= 3.32$
$5^x = 8$	$\lg$ both sides, power down, divide	$\lg 8 / \lg 5 = 1.29$
$3^{x+1} = 12$	the WHOLE bracket comes down, then -1	$2.26 - 1 = 1.26$
3,12,48,192,...	divide consecutive terms for the ratio	$\times 4 \rightarrow 768$
$\times 3$ every 20 s, after 1 min	count the STEPS 60/20 = 3	$12 \times 3^3 = 324$

PARTIAL FRACTIONS AND FACTOR THEOREM		
BOTTOM LOOKS LIKE	SET UP AS	ANSWER
$(x-1)(x+8)$ different	$A/(x-1) + B/(x+8)$	kill one bracket at a time
$(x+5)^2$ squared alone	$A/(x+5) + B/(x+5)^2$	$10/(x+4) - 5/(x+4)^2$
$(x+6)(x-1)^2$	$A/(x+6) + B/(x-1) + C/(x-1)^2$	$8/(x+6) - 1/(x-1) + 4/(x-1)^2$
$x(x^2+x+7)$ will not factorise	$A/x + (Bx+C)/(\text{quadratic})$	$4/x + (3-x)/(x^2+x+7)$
$(2x-3)$ in a bracket	kill it with $x = 1.5$, not 3	$4/(2x-3) - 1/(x-1)$
$(t-5)(t+1)$ with a story	ignore the story entirely	$A = 30, B = 4$
"is $(x-2)$ a factor"	sub $x = +2$. $f(a)=0$ means yes	$f(2)=0 \rightarrow$ yes
" $(x-3)$ is a factor, find a"	set $f(3) = 0$, solve	$9+3a-15=0 \rightarrow a = 2$
Method: $\times$ by the whole bottom, then pick the x that kills a bracket. When that runs out, compare the number of x^2 or x on each side.		

ORDER OF OPERATIONS, AND EQUATIONS		
LOOKS LIKE	MOVE	WORKED
the order itself	Brackets $\rightarrow$ pOwers $\rightarrow$ Divide and Multiply, left to right $\rightarrow$ Add and Subtract, left to right. The power outranks the divide, and the divide outranks the add.	$6 + 3 \times 10 = 36$ not 90 $5 + 2^2 + 4 - 1 = 6$
$3x - 7 = 6$	undo in reverse order	$x = 13/3 = 4.33$
$4x+2 = 5x+10$	collect x on the side that keeps it +ve	$x = -8$
$7t - (3 - t) = 2$	the minus flips EVERY sign inside	$8t-3=2 \rightarrow t = 5/8$
$(9+5k)/4 - 3 = 2k$	multiply EVERY term by 4, both sides	$k = -1$

ORDER OF OPERATIONS, AND EQUATIONS		
LOOKS LIKE	MOVE	WORKED
$5x+2y=13, x+2y=9$	a term matches $\rightarrow$ subtract	$x=1, y=4$
$2a+5b=10, 3a-4b=15$	nothing matches $\rightarrow$ scale both, then add	$a=5, b=0$
"use substitution"	make the lone letter the subject, sub in	$x=-0.5, y=-6.5$
$-5c - 7 > 8$	$\div$ by a NEGATIVE turns the sign round	$c < -3$
$34 > p - 12$	flip the whole statement to put p on the left	$46 > p \rightarrow p < 46$
"between \$199 and \$299, \$26 off"	double inequality, same op on all three	173 to 273
Number line: hollow = not included, solid = included.		

- BEFORE YOU HAND IT IN
- Calculator in **DEG?** (radians only where there is a π)
 - Every trig equation — did I give **both** answers?
 - tan second answer is **+180**, not 360 minus
 - Minus in front of a bracket — did every sign inside flip?
 - Completed square with a number in front — did it multiply what I subtracted?
 - Turning point — did I sub back into the **original**?
 - 2nd derivative: +ve is a MINIMUM
 - Did it want x, or the height / area / time?
 - Indefinite integral — **+ c?** Definite — no c, upper minus lower?
 - Area came out negative $\rightarrow$ I subtracted backwards, or it dips below the axis
 - Line of symmetry written as **x = ...**, not a bare number
 - 2 d.p. unless told otherwise. Units where asked.
 - Anything blank? Write the first line anyway. It earns.

FIRST ORDER DIFFERENTIAL EQUATIONS — MATCHING ONLY

THE STORY	EQUATION	SOLUTION AND GRAPH
decays, falls towards nothing	$dx/dt = -kx$	Ae^{-kt} . drops, flattens onto 0
grows without limit	$dx/dt = kx$	Ae^{kt} . rises ever more steeply
cools towards a room at c	$dx/dt = -k(x-c)$	$c + Ae^{-kt}$. levels onto the line $x=c$

The give-away: rate depends on the amount *itself* $\rightarrow kx$. On a **difference** from a fixed level $\rightarrow -k(x-c)$.

Constants, in this order: k from the equation using the given rate; then A from $t = 0$ in the solution. $e^0 = 1$.

coffee 90°, room 20°, $k=0.01 \rightarrow T = 20 + 70e^{-0.01t}$

MCQ tie-break: a $-k$ in the equation MUST give a $-kt$ in the power. The trap option pairs a right equation with a mismatched solution.

"Time to reach room temperature" never happens — it is asymptotic. Say "approaches 20°; from the graph about 450 to 500 min".

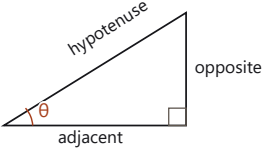
decay $dx/dt = -kx$ Ae^{-kt}

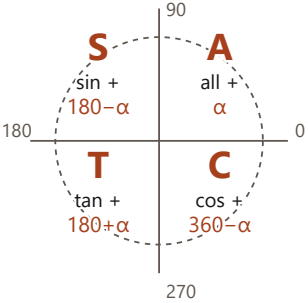
growth $dx/dt = kx$ Ae^{kt}

to a limit $-k(x-c)$ $c + Ae^{-kt}$

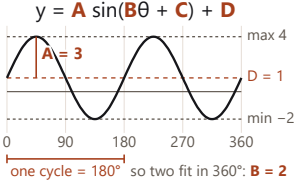
Ends at **nothing, nowhere, or flat on a line.** That one question picks the row.

TURNING POINTS, AND WORD PROBLEMS		
STEP	MOVE	BOTH EXAMPLES
1 differentiate	the lone number disappears	$y=5+6x-3x^2 \rightarrow dy/dx = 6-6x$ $y=7x^2+14x-12 \rightarrow dy/dx = 14x+14$
2 set it to zero	solve, this gives x, and only x	$6-6x = 0 \rightarrow x = 1$ $14x+14 = 0 \rightarrow x = -1$
3 find the y	sub x into the ORIGINAL , never into the derivative	$5+6-3 = 8 \rightarrow (1, 8)$ $7-14-12 = -19 \rightarrow (-1, -19)$
4 which kind is it	differentiate again. that second answer is written y" . positive = MINIMUM, negative = MAXIMUM (positive is a cup, it holds water, so it is the bottom)	$y'' = -6 \rightarrow$ MAXIMUM $y'' = 14 \rightarrow$ MINIMUM
free check, quadratics only	your formula sheet gives the turning point as $(-b/2a, c-b^2/4a)$. Same answer, five seconds, no calculus	use it to check, not to replace
"time to reach maximum height"	work the number out BEFORE differentiating	$100 \sin 30 = 50$, then $h = -5t^2 + 50t \rightarrow t = 5$ s
"largest area that can be fenced"	x is not the answer. Put x back and give the AREA	$x = 5$, but the answer is 50
"rate of change at $t = 3$ "	differentiate, then put 3 in. A negative answer is a real answer, not a mistake	-0.02

TRIG — WHICH RATIO OR RULE?		
YOU ARE GIVEN	USE	WORKED
right angle, hyp given, want the side OPPOSITE	SOH: opp = hyp × sin	40°, hyp 8 → h = 8sin40 = 5.14
right angle, adj given, want the HYPOTENUSE	CAH rearranged: hyp = adj ÷ cos	30°, adj 15 → k = 15/cos30 = 17.32
right angle, angle + one leg, no hyp anywhere	TOA: opp = adj × tan	12 tan35 = 8.40
two sides, angle unknown	inverse: sin ⁻¹ (o/h), cos ⁻¹ (a/h), tan ⁻¹ (o/a)	3/cos45 = 4.24 m
ladder, angle with the GROUND, foot 3 m out	ground = adjacent, ladder = hypotenuse	
...“angle with the WALL”	no trig at all: 90 – that	→ 45°
pendulum L, angle from the VERTICAL	rise = L(1 – cosθ). check where the angle is marked	45(1–cos30) = 6.03 cm
NO right angle: 2 angles + a side, or an angle and its opposite side	SINE RULE (not on sheet) a/sinA = b/sinB	C=35, c=40, b=50 → B = 45.80°
NO right angle: 3 sides, want an angle	COSINE RULE (not on sheet) cosA = (b²+c²–a²)/2bc	42, 37, 26 → A = 60.63°
NO right angle: 2 sides + the angle BETWEEN	COSINE RULE: c² = a²+b²–2ab·cosC	60°, 8, 6 → c = 7.21
Which rule? An angle opposite a known side, or two angles → SINE. Three sides, or two sides with the angle wedged between → COSINE. In the cosine rule the side you square on the left is the one FACING the angle.		
		
SOH sin = O/H CAH cos = A/H TOA tan = O/A opposite is ACROSS from θ. the hypotenuse is always the long sloping one.		

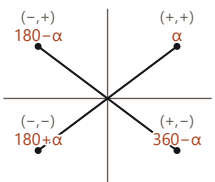
TRIG EQUATIONS — ALWAYS TWO ANSWERS		
YOU HAVE	QUADRANTS	THE TWO ANSWERS
sin = +	1st, 2nd	α and 180 – α
sin = –	3rd, 4th	180 + α and 360 – α
cos = +	1st, 4th	α and 360 – α
cos = –	2nd, 3rd	180 – α and 180 + α
tan = +	1st, 3rd	α and α + 180 – NOT 360–α
tan = –	2nd, 4th	180 – α and 360 – α
α = inverse of the value with its minus sign removed . Always 0 to 90. 4sinθ+5=3 → sin=–0.5, α=30 → 210, 330 4cosθ+2=5 → cos=0.75, α=41.41 → 41.41, 318.59 2tanθ–2=7 → tan=4.5, α=77.47 → 77.47, 257.47		
If the bracket holds 2θ: solve for 2θ, get the two answers, add 360 to each , then divide by 2. Two cycles means four answers. Both sin and cos in one equation → collapse to one function with an identity (they are on your sheet) first.		
		

WAVES — THE FOUR LETTERS		
WANTED	MOVE	WORKED
D the middle line	max and min, added, then halved	(4 + (–2)) ÷ 2 = 1
A the height	max and min, subtracted, then halved	(4 – (–2)) ÷ 2 = 3
B the squeeze	how many whole cycles fit into 360°	2 cycles → B = 2
sin or cos, and the sign in front of A	look at what the curve is doing at θ = 0: on the middle line going UP → + sin on the middle line going DOWN → – sin sitting at the MAXIMUM → + cos sitting at the MINIMUM → – cos	rising at 0 → + sin

WAVES — THE FOUR LETTERS		
WANTED	MOVE	WORKED
C the sideways shift	opposite sign to the shift. A shift right of 10 gives C = –10	no shift → C = 0
		
C is the one you cannot see here. It slides the wave sideways, and takes the opposite sign.		

WAVES — FOUR BUILT, AND THE TWO TRAPS		
GIVEN	D, THEN A, THEN B	ANSWER
max 4, min –2, 2 cycles	D = (4–2)/2 = 1 A = (4+2)/2 = 3 B = 2	y = 3sin(2θ) + 1
max –1, min –3, 2 cycles, starts mid rising	D = (–1–3)/2 = –2 A = (–1+3)/2 = 1 B = 2 rising → +	y = sin(2θ) – 2
max 6, min –2, 3 cycles, starts mid falling	D = (6–2)/2 = 2 A = (6+2)/2 = 4 B = 3 falling → –	y = –4sin(3θ) + 2
max 2, min –8, 4 cycles, starts at the min	asked with K, L, M: y = K cos(Lθ) + M. Same three numbers, different letters. M = –3 size of K = 5 L = 4 at the min → –	K = –5, L = 4, M = –3
“0.5 cycles in 180°”	B always counts cycles per 360° , so half a cycle in 180 is one whole cycle in 360	B = 1, not 0.5
“a right shift of 10°”	C takes the opposite sign	C = –10, not +10
“maximum value of...”	max = A + D. B and C make no difference to it	3sinθ–4 → 3–4 = –1
“minimum value of...”	min = – A + D. Use the SIZE of A, ignore its sign	–2cos(3θ)+5 → –2+5 = 3

VECTORS		
LOOKS LIKE	MOVE	WORKED
“magnitude”	√(x²+y²) — not on sheet. signs vanish	(–7, –11) → √170 = 13.04
“direction”	α = tan ⁻¹ (y / x), then place it!	see the quadrant box
p + q, p – q	row by row. same dimension only	(4, 12)+(2, –4) = (6, 8)
r – 3s	multiply FIRST, then subtract	→ (1, –9, –13)
(5,y)+(7,2) = 3(2x,2)	expand the scalar first → (6x,6)	x = 2, y = 4
“column vector AB”	B – A. head minus tail	A(2, 7, 2) B(8, 4, 1) → (6, –3, –1)
PQ and QP	PQ = OQ – OP. QP is every sign flipped	(–5, 0) and (5, 0)
BX + XC	middle letters MATCH → add	→ (–1, 11), BC = 11.05
PX and CX	both end at X → PX – CX	→ (5, 1), PC = 5.10
“angle between”	dot, both magnitudes, then cos ⁻¹ . formula on sheet	(12, 5)(3, 7) → 44.18°
...dot comes out negative	that means the angle is OBTUSE. keep the minus	→ 139.40°
“are they perpendicular?”	dot = 0. no cos ⁻¹ needed	
a × b	formula on sheet. middle component reversed. check: dot with either gives 0	(8, 1, 2)×(2, 1, 5) → (3, –36, 6)

VECTOR DIRECTION — PLACE THE ANGLE		
		
α = tan ⁻¹ (y ÷ x), always 0 to 90. Sketch the arrow first, then place it.		

MATRICES		
LOOKS LIKE	MOVE	WORKED
“state the order”	rows × columns. two numbers, never multiplied out	[1 4 –2] → 1×3
“order of the product, if applicable”	INNER numbers must match. OUTER numbers give the answer	(3×1)(1×4) → 3×4

MATRICES		
LOOKS LIKE	MOVE	WORKED
...when they do not match	say not possible — that IS the answer	(2×2)(1×3) → undefined
A + B, A – B	same order only. position by position	→ [[1, –2], [4, –3]]
kA	every entry, including zeros and negatives	0.5[[3, –4], [0, 2]] → [[1.5, –2], [0, 1]]
row × matrix	row of the left against column of the right, one at a time	[1 –2 5][–] → [5 –3 23]
2×2 × 2×2	layout on sheet. AB ≠ BA	→ [[29, –3], [38, –11]]
“= (m, 1), find m”	only compute the row you need; the given entry checks it	4(8)+2(9) → m = 50
“express in matrix form”	line the variables up in the same order. write 0 for any missing one	
determinant	ad – bc. formula on sheet. watch double negatives	(–2)(–4)–(2)(3) = 2
inverse	swap a and d, negate b and c , ÷ det. det=0 → none	
“solve using the inverse”	U = M⁻¹V. the inverse goes on the LEFT	→ x = –39, y = 19

DIFFERENTIATE — READ THE FUNCTION		
LOOKS LIKE	MOVE	WORKED
ax ⁿ	power to the front, then n – 1	4x* → 20x*
7x	the x vanishes	→ 7
a bare number	goes to nothing	98 → 0
6/x, 5/(12x²)	REWRITE as a negative power	6x ⁻¹ → –6x ⁻²
√x, ∛x	REWRITE as a fraction power	x ^{0.5} → 0.5x ^{-0.5}
()() two brackets	EXPAND. there is no product rule	(3x²+5)(x²+1) → 12x²+16x
(sum)/(one term)	SPLIT term by term, then rewrite each	(x²+2x+5)/x → split first
A sin x, sin Bx, cos Bx, ae ^{bx+b}	on the sheet. copy it	12cos(x–3) → –12sin(x–3)
9 – cos5x	cos → –sin, and minus a minus is plus	→ +5sin5x
A sin(Bx+C)	not on sheet: AB cos(Bx+C)	3sin(2x–7) → 6cos(2x–7)
ln(anything)	not on sheet: inside' OVER inside	ln(x²+4x) → (4x²+4)/(x²+4x)
() ⁿ a bracket to a power	CHAIN: n()^{n–1} × inside'	(x²–4) ⁵ → 10x(x²–4) ⁴
√(bracket)	rewrite as () ^{0.5} , then chain	→ 1.5x²/√(x²+2)
d²y/dx²	differentiate your own answer again	

INTEGRATE — READ THE INTEGRAND		
LOOKS LIKE	MOVE	WORKED
ax ⁿ	add 1 to the power, divide by the NEW power	–5x² → –5x²/3 + c
a bare number	times x	50 → 50x + c
1/x³, 1/x ⁿ	REWRITE as x^{–n}	→ –1/(2x²) + c
1/x	the one the power rule fails on	→ ln x + c
a/(bx+d)	NOT on sheet: (a÷b)·ln(bx+d)	7/(3–4x) → –1.75ln(3–4x)+c
√x	rewrite x ^{0.5}	→ (2/3)x ^{1.5} + c
a sin kx, a cos kx	on the sheet. DIVIDE by k	sin9x → –cos9x/9 + c
a sin(Bx+C)	not on sheet: –(a/B)cos(Bx+C)	4sin(5x+7) → –0.8cos(5x+7)+c
sin x · cos x	REWRITE with sin2x = 2sinxcosx	4sinxcosx → –cos2x + c
ae ^{kx+b}	on the sheet. divide by k	e ^x → e ^x /4 + c
a sum of terms	one at a time, ONE +c at the end	
dy/dx AND a point given	integrate, sub the point, find c, write the WHOLE equation	c = –8.25
numbers on the f	no +c. upper minus lower	f ₁ *3x = 54–1.5 = 52.5
“area bounded by...”	it is the definite integral	5x²–2x+8, 1 to 3 → 108
curve dips BELOW the axis	split, drop the minus, then add	x² from –3 to 2 → 24.25

